Cristiano Ronaldo’s Champions League Legacy: Exploratory Data Analysis

CSE 163 | Project Part 2 | November 12, 2026
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Public team GitHub: https://github.com/yiranli7-code/CSE163-UCL-Data-Analysis

1. Summary of Research Questions

RQ1 — Career contributions: Compare Cristiano Ronaldo with Lionel Messi, Robert Lewandowski, Karim Benzema and Kylian Mbappé using UEFA-listed goals, assists and G+A totals. RQ2 — Efficiency: Examine goals and goal contributions per appearance only when compatible appearance denominators are available.

RQ3 — Seasonal peaks: Compare four selected Ronaldo seasons with standout seasons from the other players; avoid making up historical assists. RQ4 — Match prediction: Use regularized logistic regression and tuned random forests to predict whether a match has at least three total goals from competition stage and each side’s pre-match scoring form.

2. Motivation

Career accumulation, exposure-adjusted rates and exceptional seasons express different kinds of football performance. Our goal is a transparent comparison, not a predetermined “greatest player” conclusion. The match-level model is a separate statistical-learning application, not evidence of Ronaldo’s individual causal impact on Real Madrid outcomes.

3. Datasets and Provenance

UEFA career snapshot: five selected players × four source attributes (player, goals, assists, appearances). UEFA selected peaks: nine player-season entries × four fields (player, season, goals, source_note). OpenFootball: 13 raw cl.txt season files (2011/12–2023/24) from the uploaded CC0 GitHub archive, converted into 1,619 main-competition matches × eight source fields. The OpenFootball match records do not contain complete player assists or minutes and are not substituted for UEFA’s player ranking tables.

4. Method

We parsed source scorelines and stage headings, excluding qualifying fixtures and shootout conversion scores while retaining extra-time goals. Every season’s record count was checked against its own # Matches header. The parser also recognizes Gruppe headings in three seasons. Pandas then validated dates and unique fixtures, calculated total goals and a binary ≥3-goal indicator, and constructed prior-date goals-for/against averages separately for each club and season; same-day scores cannot leak into features. Summary statistics, eight figures, hypothesis tests and machine-learning evaluations were computed in Python.

5. EDA Results: Size, Missingness and Variables

Career table: 5 × 4 raw and 5 × 7 with derived G+A and rate variables. The five player names are categorical (each appears once). Two appearance entries are missing (Lewandowski and Mbappé; 40%), and therefore two observations are missing in each per-appearance field (40%). Goals, assists and contributions have no missing values.

Variable	n	Mean	SD	Min	Q1	Med	Q3	Max
Goals	5	108.60	27.43	74.00	90.00	109.00	129.00	141.00
Assists	5	31.80	8.79	22.00	26.00	29.00	40.00	42.00
Appearances	3	167.33	17.90	152.00	157.50	163.00	175.00	187.00
G+A	5	140.40	34.68	100.00	119.00	131.00	169.00	183.00
Goals/app	3	0.71	0.11	0.59	0.67	0.75	0.77	0.79
G+A/app	3	0.93	0.13	0.78	0.88	0.98	1.01	1.04

Peak-season table: 9 × 4, with no missing data by isna().sum(). Player category counts: Ronaldo 4; Mbappé 2; Messi, Benzema and Lewandowski 1 each. The nine season labels are 2007/08, 2011/12, 2013/14, 2015/16, 2016/17, 2019/20, 2021/22, 2023/24 and 2025/26 (one record each). This is a purposely selected set.

Variable	n	Mean	SD	Min	Q1	Med	Q3	Max
Season goals	9	13.33	3.32	8.00	12.00	15.00	15.00	17.00

Match table: 1,619 × 8 parsed source columns; 1,619 × 10 after total_goals and high_scoring. Each row represents one completed UCL match. Stage categories: Group 1,248; Knockout 371. There are zero missing source and derived match values according to isna().sum(), while 208 season-opening matches (12.85%) lack prior-game averages by design; 1,411 matches are eligible for modeling.

Variable	n	Mean	SD	Min	Q1	Med	Q3	Max
Home goals	1619	1.67	1.44	0.00	1.00	1.00	2.00	8.00
Away goals	1619	1.30	1.29	0.00	0.00	1.00	2.00	8.00
Total goals	1619	2.97	1.73	0.00	2.00	3.00	4.00	12.00
High (0/1)	1619	0.57	0.50	0.00	0.00	1.00	1.00	1.00

Season category counts: 2011/12, 2012/13, 2013/14, 2014/15, 2015/16, 2016/17, 2017/18, 2018/19 (125 each); 2019/20 (119); 2020/21, 2021/22, 2022/23 and 2023/24 (125 each). Team labels are identifiers, not model features.

5.1 Career Contributions and Efficiency (RQ1–RQ2)

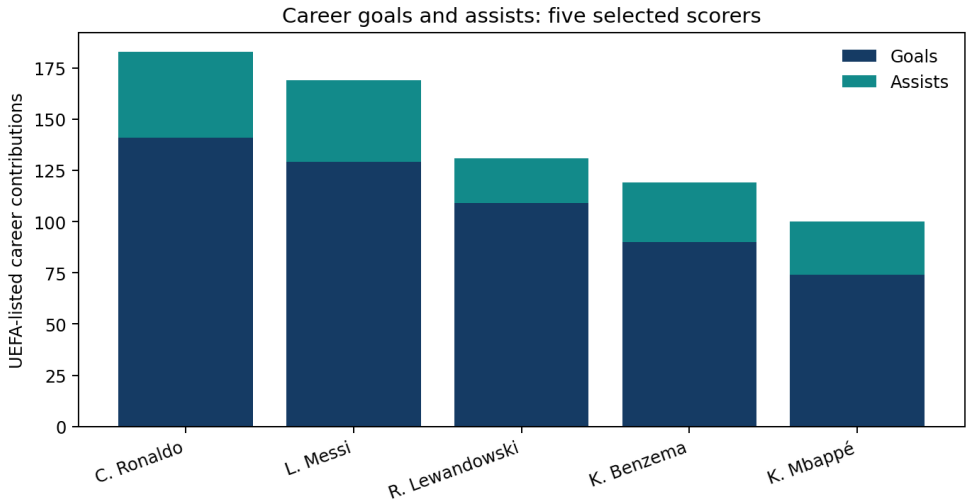


Figure 1. Ronaldo leads the five-player UEFA snapshot in listed career goals and observed goals plus assists.

The stacked bar plot separates goals and assists: Ronaldo has 141 + 42 = 183 listed contributions, compared with Messi's 129 + 40 = 169. The comparison uses one UEFA ranking convention but is not a claim about all competition-stage definitions.

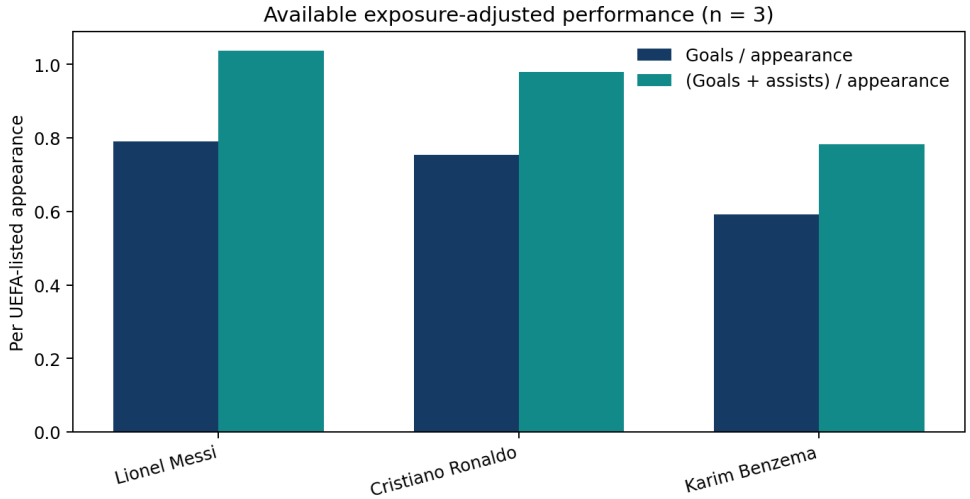


Figure 2. Messi has the highest goals per UEFA-listed appearance among the three players with observed denominators.

The grouped bars show goals/appearance and G+A/appearance. Messi has 129/163 ≈ 0.79 goals per appearance, versus Ronaldo 141/187 ≈ 0.75 and Benzema 90/152 ≈ 0.59. Missing denominators for Lewandowski and Mbappé are not zero-filled.

5.2 Selected Seasonal Peaks (RQ3)

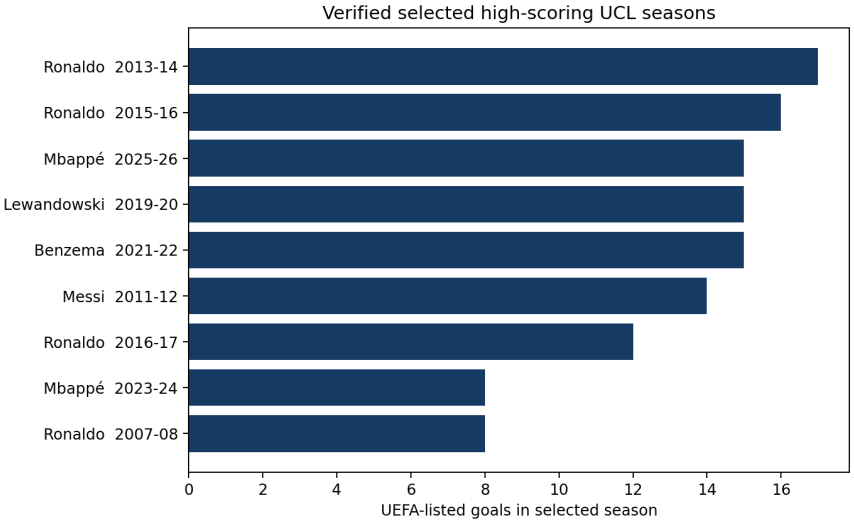


Figure 3. Ronaldo's 2013/14 total of 17 goals is the largest among the nine selected player-seasons.

A horizontal bar plot allows direct comparison of the source season-goal totals, including Ronaldo 2015/16 (16) and Benzema 2021/22 (15). The entries were deliberately selected, so this is not a distribution of all UCL player-seasons.

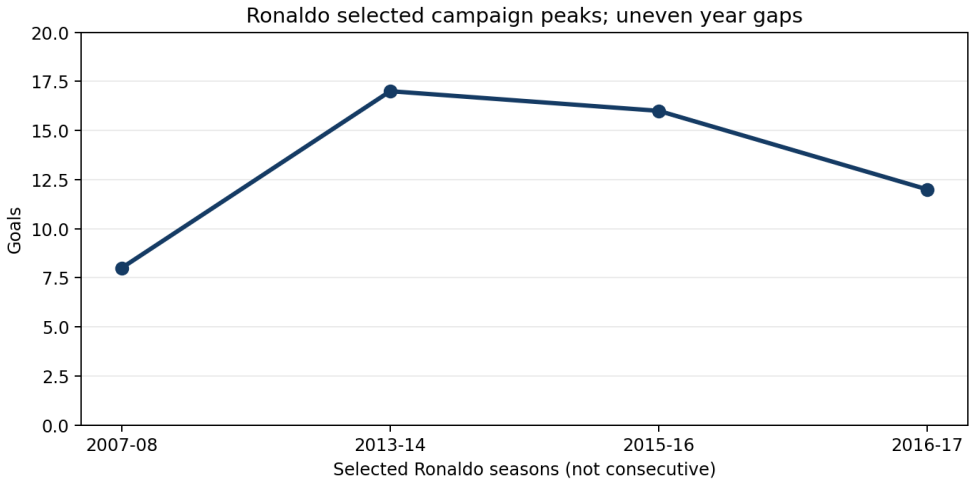


Figure 4. Ronaldo's four highlighted seasons scored 8, 17, 16 and 12 goals, respectively.

The line connects only preselected 2007/08, 2013/14, 2015/16 and 2016/17 campaigns. Uneven gaps and excluded seasons prevent a continuous career-trend interpretation; unavailable assists are not assumed to be zero.

5.3 Thirteen-Season Match Results (RQ4)

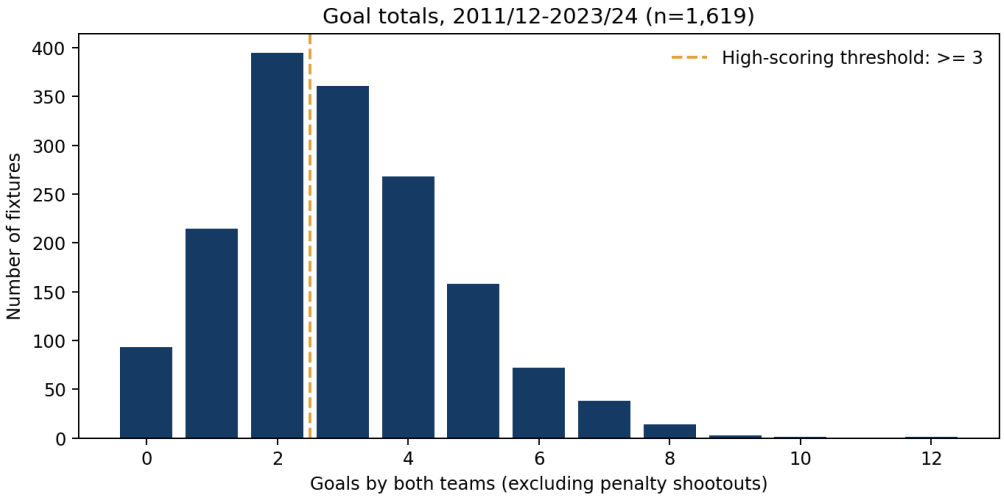


Figure 5. Across 1,619 fixtures, mean scoring was 2.97 goals per match and 916 games (56.6%) reached three goals.

This discrete goal-count plot gives the distribution underlying the binary target; totals range from 0 to 12. The dashed threshold marks three goals, and extra-time goals count but shootout conversions do not.

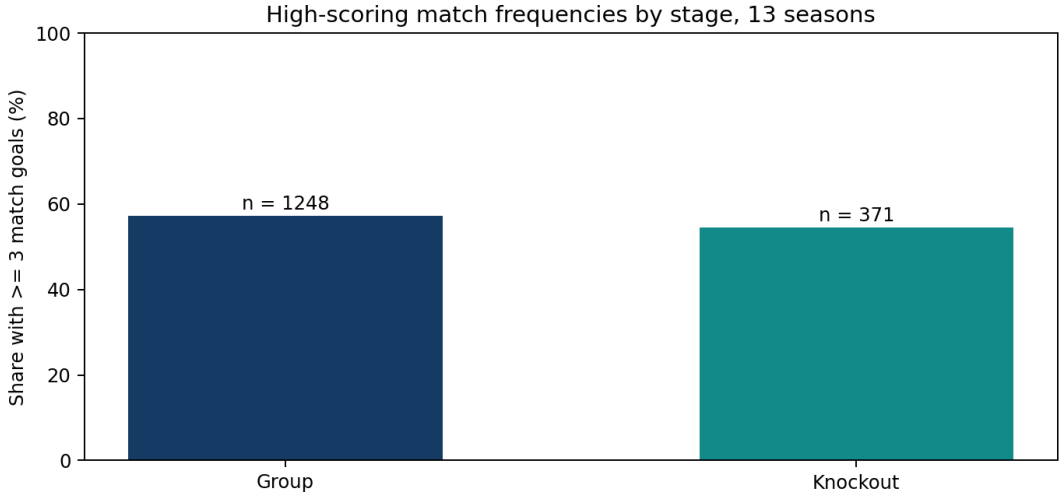


Figure 6. High-scoring proportions were 714/1,248 (57.2%) for group fixtures and 202/371 (54.4%) for knockout games.

The bar plot compares proportions rather than raw counts because the stages have unequal numbers of fixtures. The 2.76-percentage-point knockout-minus-group difference is small and descriptive; team strength and year may confound it.

5.4 Seasonal Variation and Source Audit (RQ4)

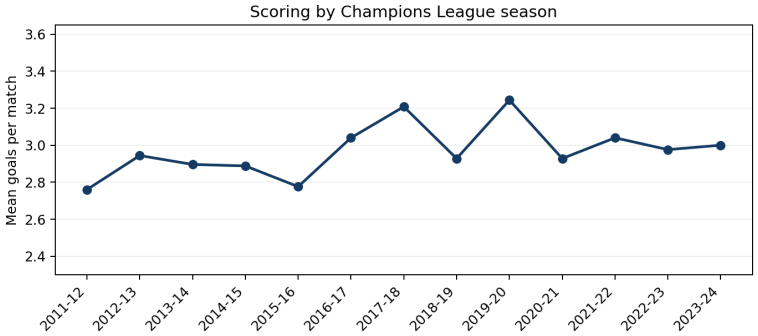


Figure 7. Seasonal average goals varied from 2.76 (2011/12) to 3.24 (2019/20), without a monotonic pattern.

The line chart highlights between-season variation in scoring opportunity and fixtures. The 2019/20 season had 119 games rather than 125, reflecting its altered knockout schedule; per-game averages remain interpretable.

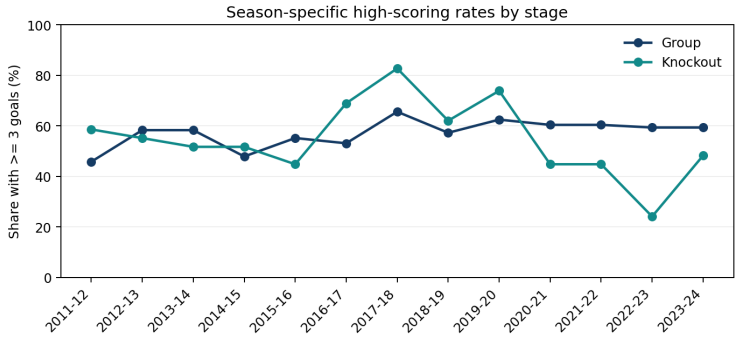


Figure 8. The high-scoring-rate gap between group and knockout games changes direction across individual seasons.

The two lines show stage-specific yearly proportions instead of obscuring heterogeneity in the pooled comparison. Their variation motivates season-level uncertainty checks. This is descriptive, not evidence that stage causes more goals.

Pre-match predictors: The four team form variables represent goals scored (GF) or conceded (GA) per prior match, home and away. Each is missing for 208 opening fixtures and available for 1,411 matches each; 1,411 matches have all four predictors.

Variable	n	Mean	SD	Min	Q1	Med	Q3	Max
Home GF	1411	1.57	0.93	0.00	1.00	1.50	2.04	6.00
Home GA	1411	1.39	0.95	0.00	0.75	1.18	2.00	7.00
Away GF	1411	1.67	0.98	0.00	1.00	1.56	2.21	7.00
Away GA	1411	1.29	0.87	0.00	0.71	1.00	1.75	6.00

6. Result Validity Challenge: Hypothesis and Assumptions

H0: Competition stage and a match's ≥3-goal indicator are statistically independent. H1: They are associated. The planned Pearson chi-square test uses the 2 × 2 stage-by-target table. Observed rows [0–2, ≥3 goals] are Group [534, 714] and Knockout [169, 202].

Assumption check: The smallest expected cell count is 161.1 (≥5), so the large-sample count criterion is met. $\chi^2(1) = 0.889$, $p = 0.346$, Cramér's $V = 0.023$. A two-sided 95% season-cluster bootstrap (3,000 draws; seed 163) estimates knockout-minus-group difference -2.76 percentage points; interval [-10.84, 4.96] points. The interval includes zero.

The strict independent-match assumption is questionable: clubs recur, matchups are strategic, and 2019/20 changed format. Resampling whole seasons partly addresses between-season heterogeneity but not within-season repeated-team dependence. Thus, the p-value is exploratory, not causal evidence or proof of equality.

7. Challenge Goals Update: Machine Learning

The four numerical predictors were pre-match goals scored and conceded averages for each team; stage was one-hot encoded. We excluded 208 opening fixtures with no within-season history. Train: 2011/12–2019/20 (n=975); validation: 2020/21–2021/22 (n=218); held-out test: 2022/23–2023/24 (n=218). The validation set selected random forest maximum depth 4; final fitting combined train and validation without using the test outcomes.

Held-out model	Accuracy	Precision	Recall	F1	ROC-AUC
Selected forest	0.546	0.542	1.000	0.703	0.473
Majority baseline	0.537	—	—	0.699	—

The tuned forest's F1 (0.703) only narrowly exceeds the always-majority-class baseline (0.699). Its ROC-AUC (0.473) is approximately chance, so these predictors have little discriminative value in the tested period. We do not claim meaningful forecasting success.

8. Work Plan Evaluation

Our proposal allocated 44 person-hours (22 per author): 4 setup, 8 cleaning/parsing, 8 historical figures, 10 modeling, 8 robustness and 6 final deliverables. The local-source parsing and validation tasks required more care than anticipated: 2019/20 had 119 games, 2020/21–2022/23 included non-English Gruppe headings, and penalty scores required special handling. Actual personal time logs were not provided; we cannot claim the original hours were met or exceeded. Estimated remaining effort: 2–3 hours each for joint source checks, interpretive review, repository publication and final presentation planning.

9. Testing and Trustworthiness

All 24 Python unittest checks pass. Tests cover seven-number summaries, missing/negative values, duplicate fixtures, expected header counts, localized stage labels, extra-time versus shootout scores, same-date no-leakage features, season resets, both bootstraps, and estimator fit/temporal partitions. We checked 12 × 125 + 119 = 1,619 games, 13 × 96 = 1,248 group games, 4,809 match goals and 916 positive target labels. All raw season files and the reproducible parser accompany this report. Run python eda.py, python model.py and python -m unittest discover -p "test_*.py". Final flake8 verification is requested on a local environment where flake8 is installed.

10. Collaboration and Consulted Resources

Authors: Rodrigo Li and Larry Zhang. As planned, Rodrigo’s focus is RQ1–RQ3 while Larry’s is RQ4; both should review interpretations and tests. Sources: UEFA historical ranking pages, OpenFootball’s CC0 repository, Python scientific-library documentation. ChatGPT was used to help draft and check code and text; both authors must verify results and follow the course’s AI disclosure requirements. No additional human collaborators are claimed.

11. Limitations and Next Steps

UEFA ranking counts can have a different competition scope from other historical articles; two player appearance denominators remain unverified. Selected player-seasons are not a full historical census. Pre-match scoring features do not fully describe opponents, lineups or team quality; repeated teams limit simple statistical independence. Next: review the raw records and league-stage format differences, check the final models jointly, and publish this version to the public team GitHub repository.

Data sources: OpenFootball Champions League (CC0) | UEFA goals | UEFA assists | UEFA appearances | UEFA season goals